Simranvir Singh Sondhi

Data Analyst | Analytics Engineer
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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Data analyst and analytics engineer who turns messy operational and regulatory data into decisions people can act on. Built a bitemporal Canadian bank regulatory reporting platform in dbt Core on BigQuery spanning 30 years of OSFI filings, governed by a BCBS 239-referenced control framework, automated with GitHub Actions CI/CD, and deployed as a live React and AWS Lambda dashboard querying the marts in production. Also delivered predictive maintenance analytics forecasting Aircraft-on-Ground risk 60-90 days ahead and benchmarked forecasting models to R2 0.98 (RMSE 3,487) on Walmart weekly sales. Works across SQL, Python, dbt Core, BigQuery, Power BI, Tableau and AWS, with a QA background that catches data defects before they reach a dashboard.

TECHNICAL SKILLS

Analytics & BI: Power BI, Tableau, Evidence (BI-as-code), Excel (Pivot Tables, VLOOKUP, Power Query), KPI reporting, dashboard design, trend and seasonality analysis, forecasting, data storytelling

Programming & Data: SQL (joins, CTEs, window functions, incremental merges), Python (Pandas, NumPy, Matplotlib, scikit-learn), dbt Core (models, snapshots, macros, tests, docs), JavaScript / Node.js, React (Vite), MySQL, Jupyter Notebook, Git / GitHub

Cloud & Pipelines: Google BigQuery, AWS (Lambda, S3, EC2, RDS, IAM, CloudWatch), Terraform, GitHub Actions CI/CD, Dagster, workload identity federation, ETL/ELT design, dimensional data modelling, automated ingestion and pipeline monitoring

Data Quality & Governance: BCBS 239 control frameworks, reconciliation and completeness controls, bitemporal / point-in-time modelling and restatement handling, data profiling and validation, anomaly detection, root-cause analysis, test planning, functional and regression testing, SDLC/STLC, audit documentation

PROFESSIONAL EXPERIENCE

Canadian Bank Regulatory Reporting Platform | dbt Core, BigQuery, Python, SQL, GitHub Actions | 2026
Live dashboard | Repository | dbt documentation

- Built an end-to-end bitemporal data platform over 30 years of OSFI bank returns (M4/P3/E3) and Bank of Canada rate data: append-only content-addressed landing zone in Python, dbt snapshots in BigQuery, and conformed core, finance and controls marts.
- Designed a BCBS 239-referenced data control framework of 21 registered controls across reconciliation, completeness, validity, timeliness and plausibility, with results logged by an on-run-end macro and aggregated into a scorecard reporting pass rate, breach count, worst-severity breach and trailing 30-day trend.
- Diagnosed a balance-sheet identity control failing 1,712 of 3,533 rows with variances up to $214B, tracing it to OSFI restructuring its own deposit line-item codes; resolved by scoping marts to a configurable rolling window while retaining full history in the snapshot.
- Caught a 2x double-counting defect hidden in subtotal validation logic by cross-checking against OSFI published validation rules, and corrected quarterly ratio math by de-cumulating year-to-date income statement figures, verified against RBC published net interest margin, ROE and efficiency ratio.
- Keyed incremental merges on knowledge time rather than business time so restatements of prior periods are never silently dropped, and proved it with a live restatement that merged exactly one row; reconciled a $187.7B gap between OSFI published industry totals and the sum of reported institutions by ingesting a second source covering foreign bank branches.
- Automated the pipeline in GitHub Actions with a scheduled daily run, slim CI on pull requests and dbt docs auto-published to GitHub Pages, modelled asset lineage in Dagster, and isolated CI into disposable schemas after finding an environment-naming defect that let unreviewed CI runs overwrite production tables.

Bank Regulatory Platform - Web Serving Layer | React, AWS Lambda, Terraform, BigQuery, Node.js | 2026
Live dashboard | Repository

- Shipped a responsive React (Vite) dashboard on Netlify backed by a Node 20 AWS Lambda Function URL in ca-central-1, exposing six endpoints over the institution, balance sheet, income statement, financial metrics and control scorecard models and querying BigQuery live on every request.
- Provisioned the Lambda, IAM role and Google Cloud workload identity federation in Terraform so the API authenticates to BigQuery with no service account key stored anywhere.
- Bound every user filter through BigQuery query parameters, validated dataset names at cold start, and capped each query with a byte ceiling so a runaway scan fails instead of billing.

- Designed mobile-first layouts that convert six-column financial tables into labelled blocks below 640px and reserve colour exclusively for control severity, keeping breaches the only signal that raises its voice.

Analize-360: Predictive Fleet Analytics for Commercial Aviation | AWS S3, Databricks, Delta Lake, Python | 2025

- Processed 20,000+ NASA CMAPSS sensor readings through a Delta Lake pipeline on AWS S3 and Databricks.
- Trained an LSTM model for Remaining-Useful-Life prediction and classified Aircraft-on-Ground risk 60-90 days ahead of failure.
- Applied genetic-algorithm maintenance scheduling targeting 99.5% dispatch reliability.

Walmart Weekly Sales Forecasting | Python, scikit-learn, Keras | 2025

- Benchmarked Linear Regression, MLP, Random Forest and deep neural network models on weekly store sales; Random Forest won at R2 0.98 and RMSE 3,487.
- Engineered lag and calendar features and enforced a chronological train-test split to prevent look-ahead leakage.

AI Demand Planner | FastAPI, React, ARIMA, XGBoost | 2025

- Built CSV ingestion with schema validation and forecast confidence bands over ARIMA and XGBoost models.
- Implemented reorder-point, safety-stock and Monte Carlo stockout simulation for inventory planning.

Canada Quality of Life Power BI Dashboard | Power BI, Statistics Canada data | 2025

- Consolidated Statistics Canada datasets across society, prosperity, health, governance and environment domains (2021-2024).
- Built composite indices, synchronized slicers and an executive summary page for non-technical readers.

EDUCATION

Graduate Certificate, Data Analytics | Humber Polytechnic, Etobicoke, ON | May 2025 - Feb 2026
GPA 88.8% (Honours)

Graduate Certificate, Information Technology | Algoma University, Brampton, ON | Sep 2024 - Apr 2025
GPA 83%

Bachelor of Computer Applications | Guru Nanak Dev University College, India | Dec 2020 - May 2023

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02) - exam preparation in progress, 2026
- Forage job simulations: BCG Data Science (churn model, 50% recall), Deloitte Australia Forensic Technology (Tableau anomaly detection), TCS Data Visualization